Referee report

Summary:

This paper first documents a positive relationship between oil price volatility and sovereign spreads, as well as a positive relationship between changes in oil prices and changes in sovereign spreads, using a quarterly panel of 35 oil-importing emerging-market economies over 1995–2019. The paper then develops a quantitative sovereign default model with long-term debt in which oil is imported as an input to domestic production and the global oil price follows a stochastic volatility process. The calibrated model is used to distinguish the effects of oil price level shocks from those of time-varying oil price volatility. Comparing the full model with versions that shut down oil price volatility or both oil price level and volatility shocks, the paper finds that time-varying oil price volatility increases average spreads, spread volatility, and default rates. In the model, higher oil price volatility raises expected future oil prices under the assumed stochastic process, which implies lower expected future oil imports and output, higher future default probabilities, lower bond prices, and higher sovereign spreads. The model further generates a precautionary-borrowing channel: in high-volatility states, particularly when oil prices are also high, the government borrows more in advance to insure against potentially higher future import costs and borrowing rates. This additional borrowing increases future debt exposure and default risk.

Major comments:

1. **More direct empirical validation of the central mechanism.**

   A central mechanism in the paper is that, under the assumed stochastic process, higher oil price volatility raises expectations of future oil prices, which in turn lowers expected future oil imports and output, increases default risk, reduces bond prices, and raises sovereign spreads. While the model delivers this transmission mechanism clearly, several of its key intermediate predictions currently receive limited direct empirical validation.

   a. Expectations. The mechanism relies importantly on the idea that higher oil price volatility changes expectations about future oil prices. It would therefore be useful to provide direct evidence on this channel. For example, oil price futures or survey-based forecasts such as Consensus Forecasts could be used to examine whether periods of higher oil price volatility are associated with higher expected future oil prices.

   b. State dependence. The model generates several testable state-dependent predictions. Figure 3 shows that the response of sovereign spreads to oil price level shocks is stronger when domestic output is low and in high-volatility states, while Figure 5 shows that this response is also amplified at higher debt levels. It would strengthen the paper to test empirical analogues of these predictions in the panel data. For example, the authors could

interact oil price changes with measures of domestic economic conditions such as GDP growth, output gaps, or recession indicators, and debt-to-GDP. These interactions would provide a closer empirical counterpart to the model's spread-slope results.

c. Inference. In Table 1, oil price changes and oil price volatility are common across countries within each quarter, while standard errors are clustered only by country. Since identification comes largely from time-series variation, country clustering may not fully account for common shocks affecting multiple countries within the same quarter. I suggest reporting two-way clustered standard errors by country and quarter.

2. **Separate the pure volatility effect from the volatility-induced increase in expected oil prices**

A central mechanism of the paper is that higher oil price volatility raises expected future oil prices, which in turn reduces expected future imports and output and raises default risk and sovereign spreads. Figure 4(d) suggests that this channel is quantitatively important. Under the assumed process for log oil prices, an increase in conditional volatility mechanically raises the conditional mean of the oil price level even when the conditional mean of log oil prices is unchanged. Thus, the benchmark specification combines a pure increase in dispersion with a volatility-induced shift in expected oil prices. Even if this expectation channel is empirically supported as discussed in Comment 1(a), it remains useful to distinguish it from the pure effect of greater dispersion.

Greater dispersion can affect default risk through changes in the probability of default even when the conditional mean of the oil price level is held fixed. I therefore think it would be useful to report at least one exercise that separates these two effects explicitly. One possible exercise would be to consider a mean-preserving specification such as: $\log p_t - \bar{p} = \rho_p (\log p_{t-1} - \bar{p}) - \frac{1}{2}[e^{2\sigma_t} - e^{2\mu}] + e_t^{\sigma_t} u_t$. This adjustment keeps the conditional expectation of the oil price level fixed at its benchmark value while allowing its conditional dispersion to vary. The authors could then recompute the responses of default probabilities, spreads, and borrowing.

This exercise would also help clarify the precautionary-borrowing mechanism in Section 6.3. Under the benchmark specification, higher volatility raises expected future oil prices, providing a direct motive to borrow in advance against higher future import costs and borrowing rates. Once this mean effect is removed, any remaining increase in borrowing would reflect a precautionary response to greater oil price risk itself rather than to a higher expected oil price level.

3. **The substitution elasticity is central to the real-side transmission mechanism**

   The quantitative transmission from oil price shocks to repayment capacity depends critically on the elasticity of substitution between the domestic input and imported oil in the CES production technology. This parameter governs how strongly oil demand responds to an increase in oil prices and therefore affects both the total value of oil imports and the decline in domestic production.

   The response of total oil import expenditures depends directly on the elasticity of oil demand. If demand is sufficiently elastic, the decline in import quantity can more than offset the increase in price, causing the total value of oil imports to fall; if demand is inelastic, it rises. At the same time, the elasticity of substitution determines the extent to which production declines as imported oil use falls. Thus, the same oil price shock can imply quite different quantitative decompositions between the import-cost and production channels.

   The paper could provide more sensitivity analysis around this parameter and, if possible, discipline it using evidence on energy demand or imported-input substitution relevant for the economies in the sample. At a minimum, it would be useful to show how oil imports, the total value of oil imports, output, default probabilities, and sovereign spreads respond over a plausible range of elasticities. Given the heterogeneity of the 35 oil-importing economies, the paper could also discuss the implications of imposing a common elasticity.

4. **What is specifically "oil" about the mechanism, and how distinct is the contribution from existing stochastic volatility sovereign default models?**

   The paper motivates its contribution in terms of oil price volatility in oil-importing economies. However, as currently modeled, oil enters as an imported intermediate input in a CES production technology, with an exogenous world price. This raises a broader question about the economic specificity of the proposed mechanism. Mathematically, the same structure appears to apply to volatility in the price of any sufficiently important imported intermediate input: a higher input price expectation reduces expected optimal input demand and production, changes the import and external financing needs, and thereby affects repayment capacity and sovereign risk.

   More broadly, while the exact combination of oil import dependence, stochastic oil price volatility, and endogenous sovereign default appears relatively novel, several closely related mechanisms have already been explored in the literature cited by the paper. For example, Seoane (2019) studies stochastic income volatility in an endogenous sovereign default model, while Johri, Khan, and Sosa-Padilla (2022) introduce stochastic volatility

in the world interest rate within a long-duration sovereign default framework and document related state dependence with respect to volatility and debt. Other related work considers oil price uncertainty in oil-importing small open economies and imported-input or terms-of-trade shocks in sovereign default models. Given these close connections, the paper would benefit from sharpening the discussion of what the oil-specific production structure adds relative to these existing mechanisms, and which quantitative implications arise specifically from introducing oil price volatility rather than from stochastic volatility in external fundamentals more generally.

One natural empirical test would be to examine whether the effect of global oil price volatility is stronger in countries with greater oil import dependence. Such cross-country exposure heterogeneity would provide direct support for the proposed oil channel. Additional robustness could control for broader imported-input or producer-price volatility to assess whether the oil volatility coefficient contains information beyond a generic global input cost shock. If feasible, a simple model-based comparison with a generic imported-input price process matched on persistence and unconditional volatility could also help assess how much of the quantitative effect is specific to oil rather than imported-input volatility more generally.

5. **The precautionary-borrowing result may depend importantly on the assumption of risk neutral lenders**

The precautionary-borrowing mechanism in Section 6.3 is one of the more distinctive implications of the paper. In high-volatility states, the government borrows more today because it anticipates that future oil prices and future borrowing costs may be higher, so current borrowing serves as a form of insurance against adverse future states. The model, however, assumes perfectly competitive risk-neutral lenders, so bond prices reflect expected default losses but not compensation for systematic risk.

It would be useful to discuss how robust this precautionary-borrowing mechanism is to lender-side risk pricing. If oil price volatility is correlated with global bad states, then default losses may occur precisely when lender marginal utility is high. With risk-averse lenders, sovereign borrowing costs would then contain an additional risk-premium component. This could materially affect the government's incentive to borrow in advance. If future risk premia are expected to rise more strongly than in the risk-neutral benchmark, the incentive to front-load borrowing could become stronger; on the other hand, if the current high-volatility state is already associated with a high risk premium, borrowing today may itself become sufficiently expensive that the precautionary-borrowing response is attenuated or even reversed.

I do not necessarily think the paper must introduce a full risk-averse lender block. However, because the precautionary-borrowing result is emphasized as an important mechanism, the paper should clarify that its quantitative strength is conditional on risk-neutral pricing and discuss whether the mechanism is likely to survive when spreads also contain compensation for systematic risk. If feasible, a simple robustness exercise with an exogenous stochastic discount factor or reduced-form global risk-premium component would be informative.

Minor comments:

1. I would avoid using "oil price uncertainty" and "oil price volatility" interchangeably. The empirical and quantitative analysis in the paper focuses on time-varying conditional volatility, whereas "uncertainty" is a broader concept that may also encompass disagreement, forecast dispersion, ambiguity, or other forms of imperfect information. I therefore suggest using "oil price volatility" more consistently throughout the paper, unless the authors intend to make a broader uncertainty claim.
2. The distinction between the exogenous endowment y and endogenous final output x could be clarified throughout the paper. At several points, the term "output" appears to be used for both objects, which makes the economic interpretation difficult to follow. This distinction also matters for the default-cost specification: default costs are tied to the exogenous endowment y, while oil price shocks affect repayment capacity primarily through endogenous production x. It would therefore be useful to assess whether this modeling choice affects the state dependence of default incentives. A robustness exercise in which default costs depend on total output or total resources would help clarify this issue.
3. The empirical regressions already control for changes in the real exchange rate. However, the mechanism may be particularly strong when an increase in oil prices coincides with a depreciation, since both the oil import and the domestic-currency burden of foreign-currency debt may deteriorate simultaneously. An interaction between oil price shocks or oil price volatility and exchange-rate depreciation, or, where available, foreign-currency debt exposure, could therefore provide a useful additional test of this amplification channel.
4. There appears to be a typo in the discussion of the CES technology on page 12 and in Table 2. For the specification $x = \left[(1-\alpha)y^{\theta} + \alpha m^{\theta}\right]^{\frac{1}{\theta}}$, the elasticity of substitution is $\frac{1}{1-\theta}$, rather than $\frac{1}{\theta}$. Since the benchmark calibration sets $\theta = 0.5$, both expressions happen to equal 2, so this does not appear to affect the reported benchmark value, but the notation should be corrected.